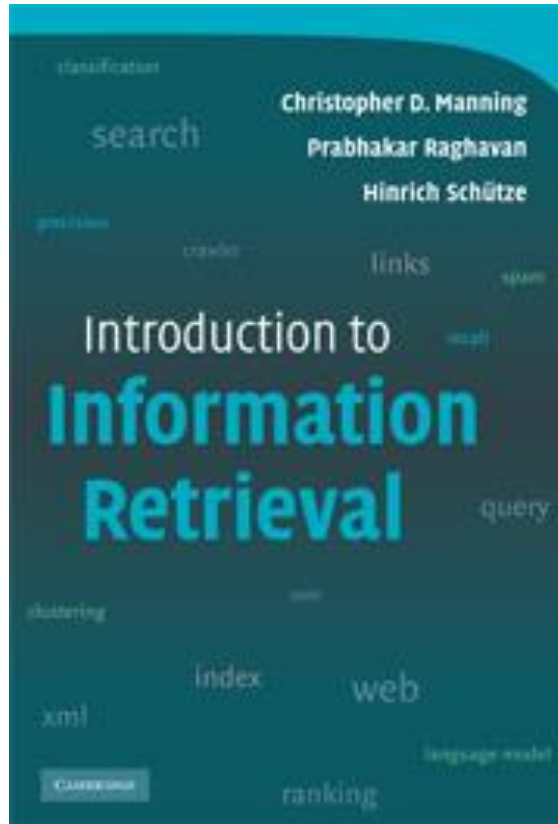


CS 419

Course Information

- This course presents an introduction to the field of information retrieval and discusses various fundamental techniques to handle information.
- Topics included:
 - The Boolean Models
 - The Vector Space Model: Term Weighting and Scoring, TF-IDF
 - Index Construction: Storing and Accessing
 - Documents Categorization and Clustering
 - Relevance Feedback and Query Expansion
 - Evaluation in Information Retrieval
 - Link Analysis (optional)
 - Visual Information Retrieval System (Image Retrieval)

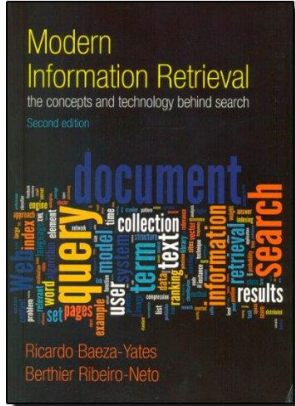
Text books



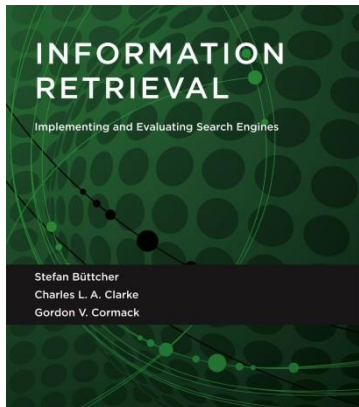
- C. Manning, P. Raghavan, H. Schütze: ***Introduction to Information Retrieval***, 2008

Available at: <http://www-nlp.stanford.edu/IR-book/>

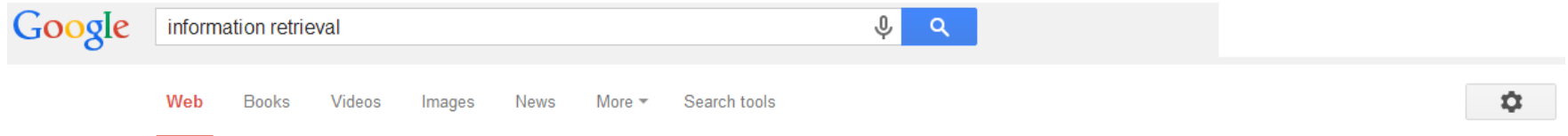
Others



- ***Modern Information Retrieval.*** Ricardo Baeza-Yates and Berthier Ribeiro-Neto, Addison-Wesley, 2011.
- ***Information Retrieval: Implementing and Evaluating Search Engines.*** Stefan Buttcher, Charlie Clarke, Gordon Cormack, MIT Press, 2010.



What is information retrieval?



About 11,200,000 results (0.31 seconds)

Information retrieval

Information retrieval is the activity of obtaining information resources relevant to an information need from a collection of information resources. Searches can be based on metadata or on full-text indexing. [Wikipedia](#)

experiments across the broad area of information retrieval. Topics of ...

Information Retrieval - Springer

link.springer.com/journal/10791 ▾ Springer Science+Business Media ▾

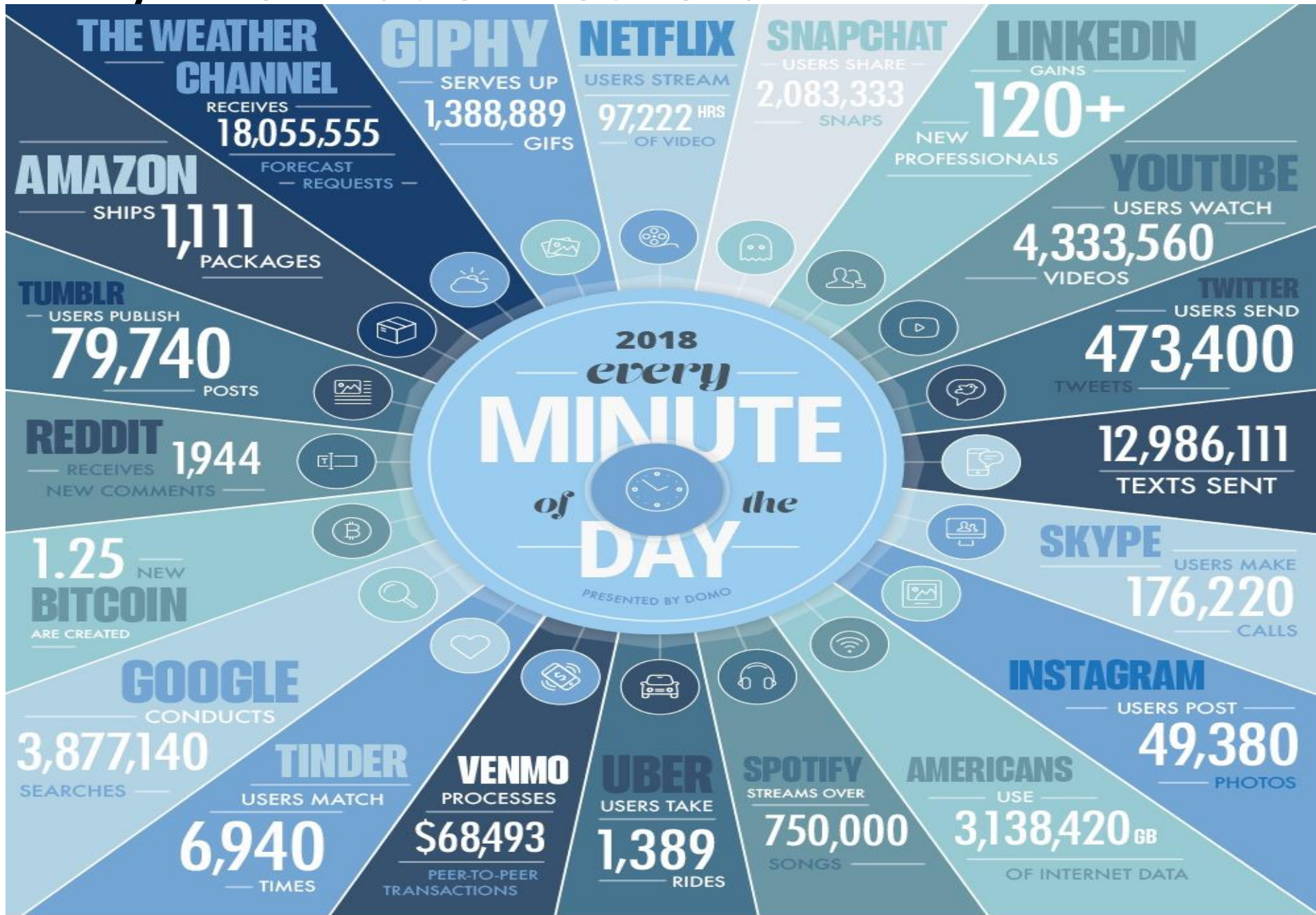
Subscription e-journal dedicated to theory and experimentation in information retrieval. Sample copy available.

Why information retrieval

- Information overload
 - “It refers to the difficulty a person can have understanding an issue and making decisions that can be caused by the presence of too much information.” - wiki



Why information retrieval ?



Why information retrieval ?

JAN
2020

DIGITAL AROUND THE WORLD IN 2020

THE ESSENTIAL HEADLINE DATA YOU NEED TO UNDERSTAND MOBILE, INTERNET, AND SOCIAL MEDIA USE

TOTAL
POPULATION



7.75
BILLION

URBANISATION:

55%

UNIQUE MOBILE
PHONE USERS



we
are
social

5.19
BILLION

PENETRATION:

67%

INTERNET
USERS



4.54
BILLION

PENETRATION:

59%

ACTIVE SOCIAL
MEDIA USERS



3.80
BILLION

PENETRATION:

49%

Why information retrieval

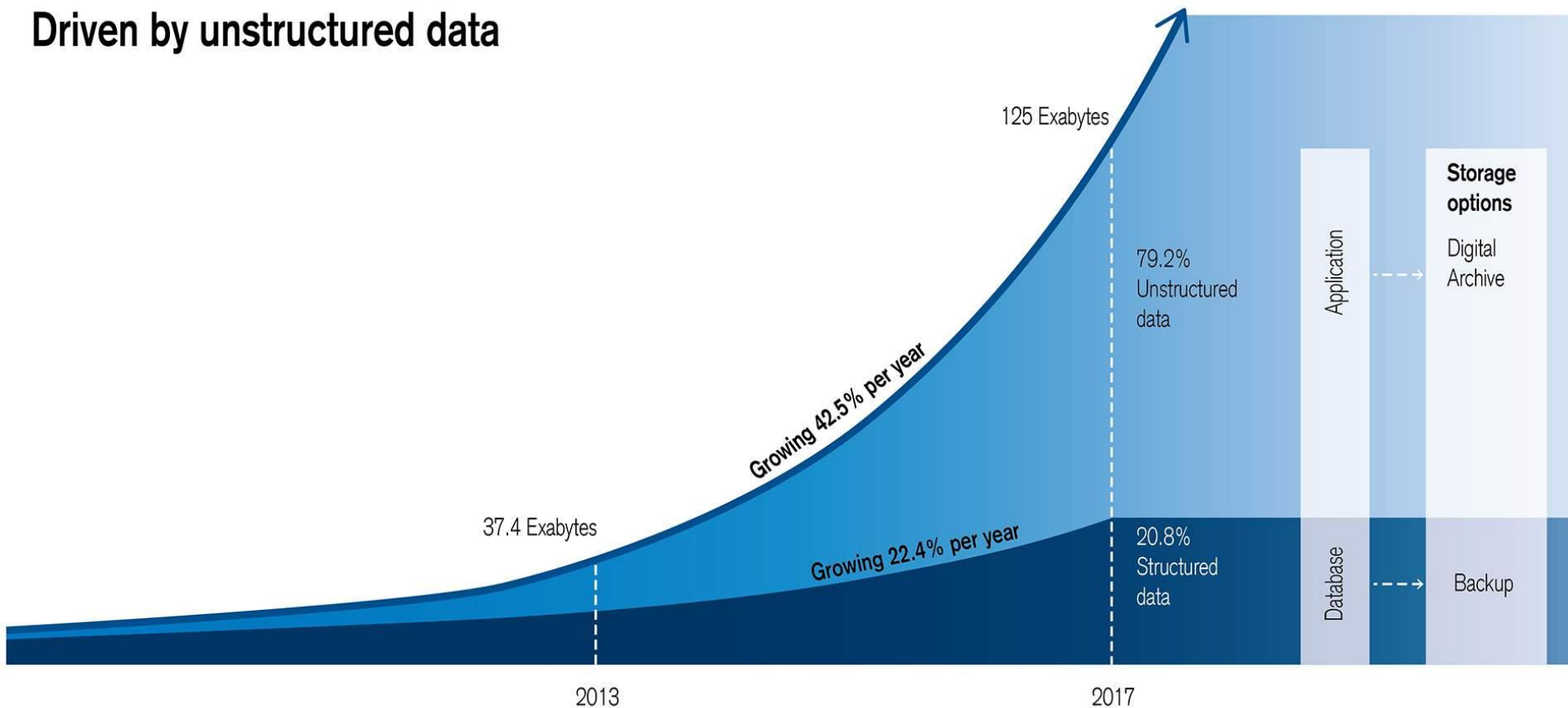
- Handling unstructured data
 - Structured data: database system is a good choice
 - Unstructured data is more dominant
 - Text in Web documents or emails, image, audio, video...
 - “85 percent of all business information exists as *unstructured data*” - Merrill Lynch
 - Unknown semantic meaning

Why information retrieval

- Handling unstructured data

Data growth

Driven by unstructured data



Why information retrieval

- What types of unstructured information exist?
 - Text: Web pages, books, articles, papers, reports, letters, blogs, ...
 - Conversational: Emails, tweets, comments, ...
 - Graphics & images, presentations
 - Speech & video
 - Maps & satellite imagery
 - Local business information, yellow pages
 - ...



Why information retrieval

- An essential tool to deal with information overload

What you
want



You are
here!

History of information retrieval

- Idea popularized in the pioneer article “***As We May Think***” by Vannevar Bush, 1945
 - “*Wholly new forms of encyclopedias will appear, ready-made with a mesh of associative trails running through them, ready to be dropped into the memex and there amplified.*” -> **WWW**
 - “*A memex is a device in which an individual stores all his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility.*” -> **Search engine**

Major research milestones

- Early days (late 1950s to 1960s): foundation of the field
 - Luhn's work on automatic indexing
 - Cleverdon's Cranfield evaluation methodology and index experiments
 - Salton's early work on SMART system and experiments
- 1970s-1980s: a large number of retrieval models
 - Vector space model
 - Probabilistic models
- 1990s: further development of retrieval models and new tasks
 - Language models
 - TREC evaluation
 - Web search
- 2000s-present: more applications, especially Web search and interactions with other fields
 - Learning to rank
 - Scalability (e.g., MapReduce)
 - Real-time search

History of information retrieval

- Catalyst
 - Industry: web search engines
 - WWW unleashed explosion of published information and drove the innovation of IR techniques
 - First web search engine: *“Oscar Nierstrasz at the University of Geneva wrote a series of Perl scripts that periodically mirrored these pages and rewrote them into a standard format.”* Sept 2, 1993
 - Lycos (started at CMU) was launched and became a major commercial endeavor in 1994
 - Booming of search engine industry: *Magellan, Excite, Infoseek, Inktomi, Northern Light, AltaVista, Yahoo!, Google, and Bing*

Global Search Engine Market



Google

93.24%

bing

3.06%

YANDEX RU

1.45%

Yahoo!

0.84%

DuckDuckGo

0.54%

Ecosia

0.31%

**Search Engine Market Share
Europe - August 2020**

Global Search Engine Market

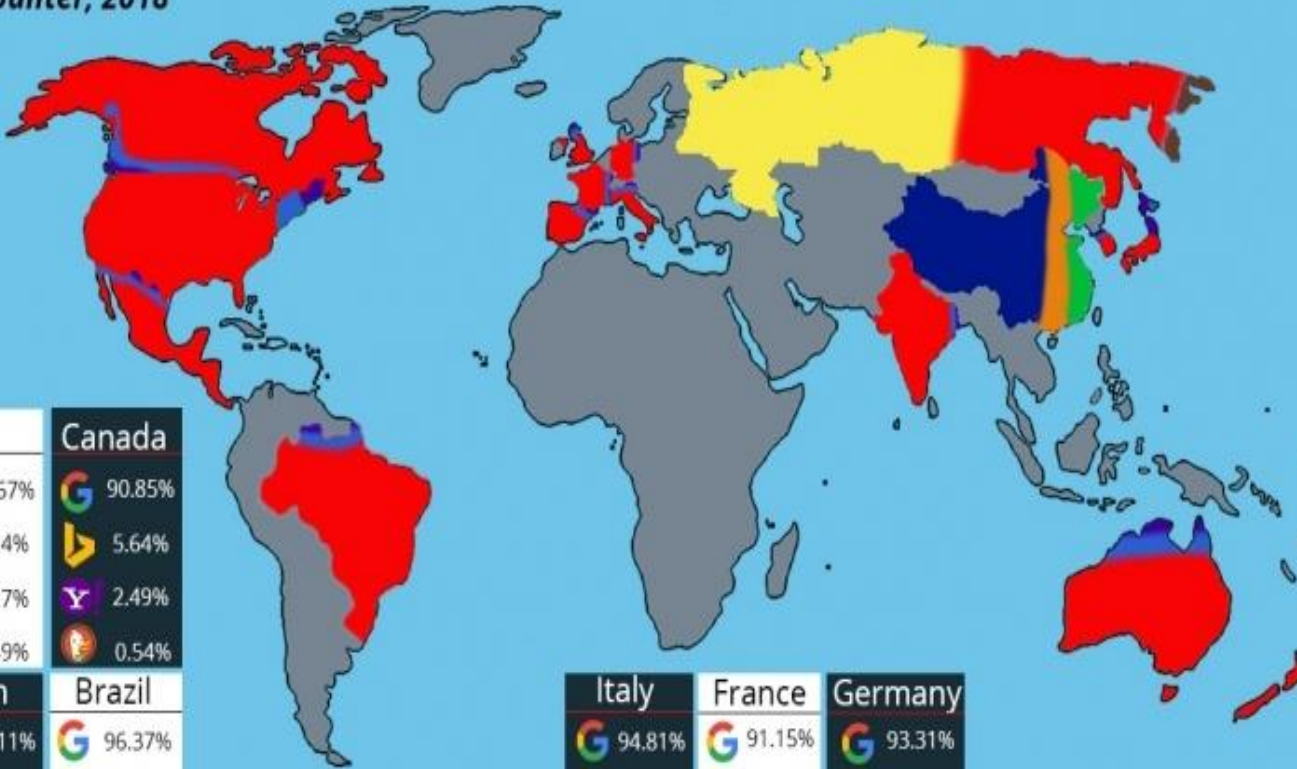
GLOBAL SEARCH ENGINE MARKET SHARE

LEGEND



Source: StatCounter, 2018

ALPHAMETIC



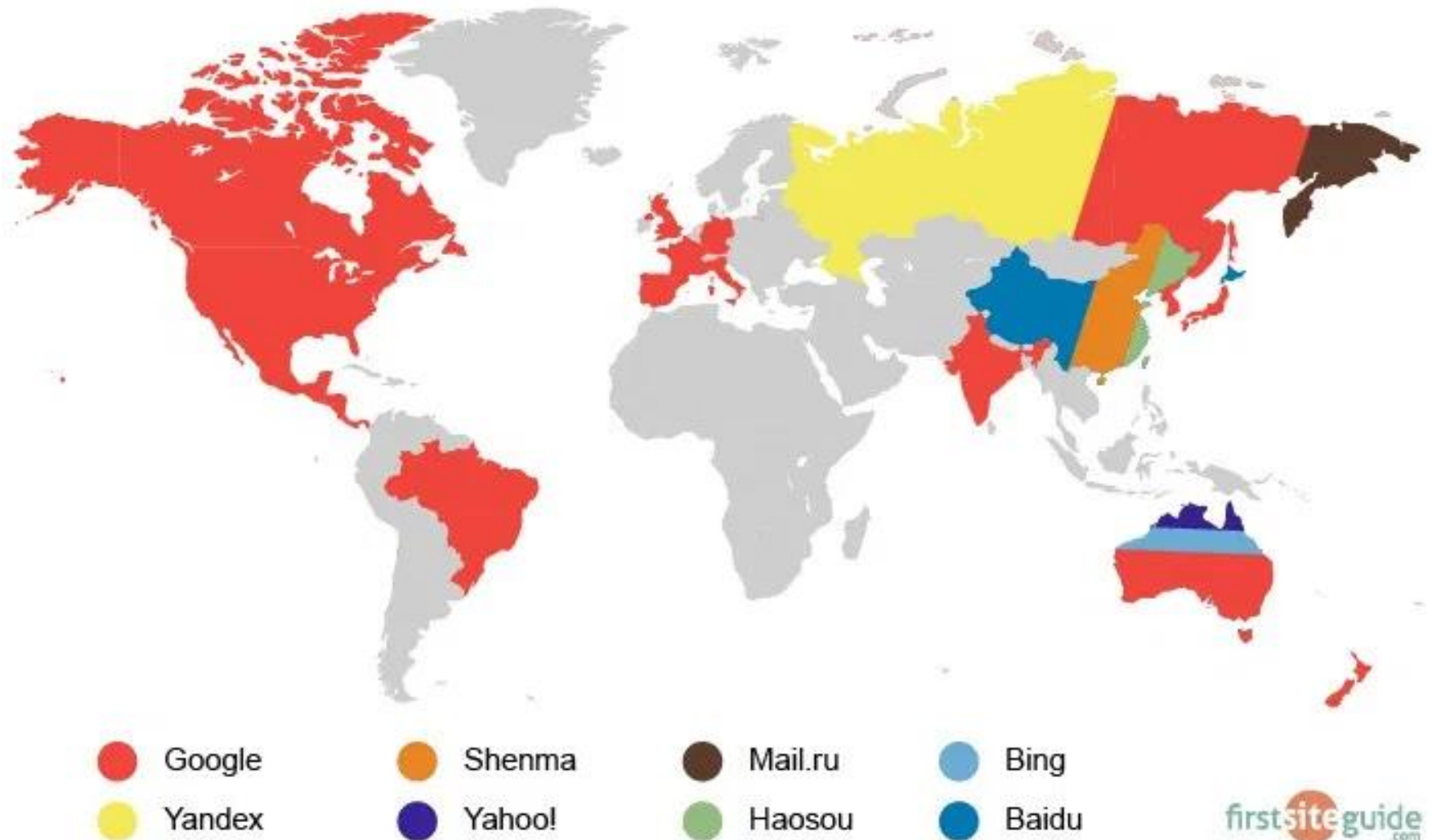
U.S.A	U.K	Canada
87.28%	89.67%	90.85%
6.91%	7.24%	5.64%
4.65%	2.27%	2.49%
0.59%	0.49%	0.54%
Mexico	Spain	Brazil
95.27%	95.11%	96.37%
3.23%	3.4%	2.02%
1.39%	1.26%	1.51%
0.05%	0.15%	0.06%

Italy	France	Germany
94.81%	91.15%	93.31%
3.52%	5.07%	4.49%
1.17%	1.92%	1.03%
0.15%	0.83%	0.51%

Russia	India
51.08%	97.35%
45.27%	1.68%
2.12%	0.91%
0.57%	0.04%
S. Korea	China
84.41%	70.26%
4.84%	19%
4.84%	4.26%
2.42%	3.71%
Australia	Japan
93.16%	70.31%
5.12%	24.03%
0.6%	4.78%
0.48%	0.58%

Global Search Engine Market

Top search engine market share statistics & facts in 2019



How to perform information retrieval

- Information retrieval when we did not have a computer



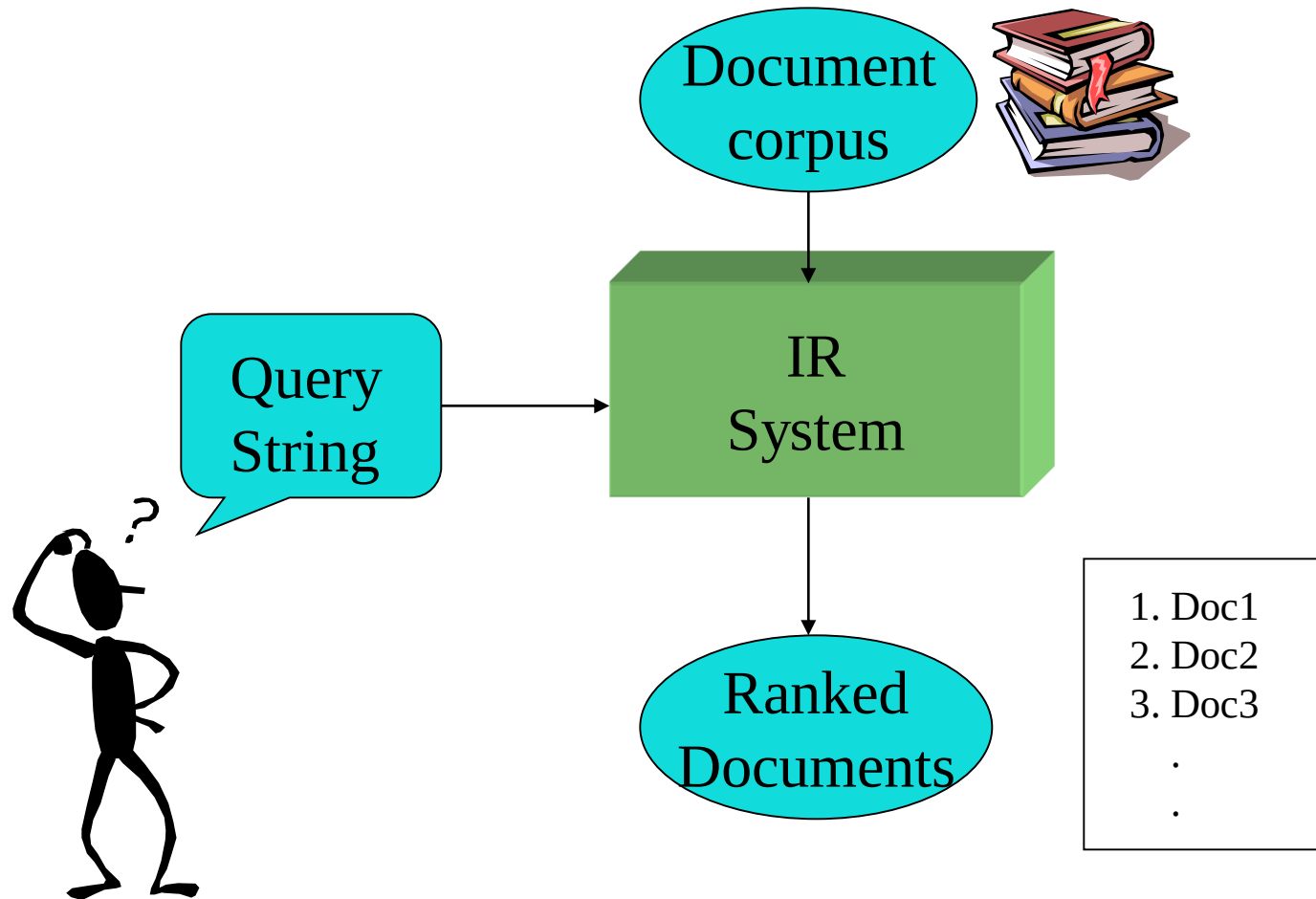
Core concepts in IR

- Query representation
 - Lexical gap: say v.s. said
 - Semantic gap: ranking model v.s. retrieval method
- Document representation
 - Specific data structure for efficient access
 - Lexical gap and semantic gap
- Retrieval model
 - Algorithms that find the most relevant documents for the given information need

Typical IR Task

- Given:
 - A corpus of textual natural-language documents.
 - A user query in the form of a textual string.
- Find:
 - A ranked set of documents that are relevant to the query.

IR System



Relevance

- Relevance is a subjective judgment and may include:
 - Being on the proper subject.
 - Being timely (recent information).
 - Being authoritative (from a trusted source).
 - Satisfying the goals of the user and his/her intended use of the information (*information need*).

A glance of modern search engine

- In old times



A glance of modern search engine

The image shows a Google search interface for the query "uva". The search bar is highlighted with a red dashed box and labeled "Demand of understanding". Below the search bar, the navigation tabs (Web, Maps, Images, News, Shopping, More, Search tools) are visible. The search results for "uva" are displayed, with the top result being "The University of Virginia". This result is highlighted with a red dashed box and labeled "Demand of efficiency". The result snippet includes the URL "www.virginia.edu/", the name "University of Virginia", and a brief description: "The University of Virginia in Charlottesville, VA was founded in 1819 by Thomas Jefferson. The". Below the snippet, there are four star ratings (4.9 stars) and a link to "University of Virginia". To the right of the result, there is a map of the University of Virginia campus, which is also highlighted with a red dashed box and labeled "Demand of convenience". The map shows the location of "The Rotunda" and "University Ave". Below the map, there is a "Directions" button. The search results are also annotated with "Demand of accuracy" and "Demand of diversity". The "Demand of accuracy" is indicated by a red dashed box around the search results area. The "Demand of diversity" is indicated by a red dashed box around the "Images for university of virginia" section, which shows four small images of the university campus. The "Demand of efficiency" is indicated by a red dashed box around the top result snippet. The "Demand of convenience" is indicated by a red dashed box around the map and directions section. The "Demand of accuracy" is indicated by a red dashed box around the search results area. The "Demand of diversity" is indicated by a red dashed box around the "Images for university of virginia" section.

Google

uva

Demand of understanding

Web Maps Images News Shopping More Search tools

About 103,000,000 results (0.65 seconds)

Demand of efficiency

Demand of convenience

The University of Virginia
www.virginia.edu/ University of Virginia
The University of Virginia in Charlottesville, VA was founded in 1819 by Thomas Jefferson. The
4.9 ★★★★★

University of Virginia
en.wikipedia.org
The University of Virginia is a research university.

University of Virginia
colleges.usnews.com
Is University of Virginia a good school?

VIRGINIAS
www.virginia.edu
The University of Virginia Network. The network

Images for university of virginia Report images

More images for university of virginia

Mascot: University of Virginia Cavalier
Founder: Thomas Jefferson
Founded: 1819, Charlottesville, VA
Colors: Blue, Orange

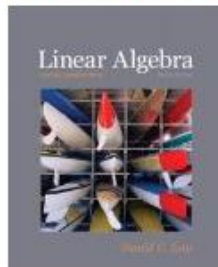
Recent posts

#UVA's Center for Politics and Politico have teamed up to offer interactive election ratings. #politics #elections #voting 1 hour ago

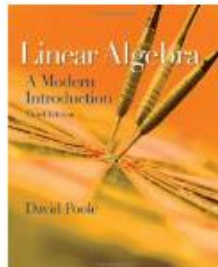
IR is not just about web search

- Web search is just one important area of information retrieval, but not all
- Information retrieval also includes
 - Recommendation

Recommended Based on Your Browsing History



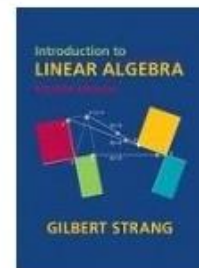
Linear Algebra and Its Applications...
› David C. Lay
Hardcover
★★★★☆ (84)
~~\$183.33~~ **\$141.16**



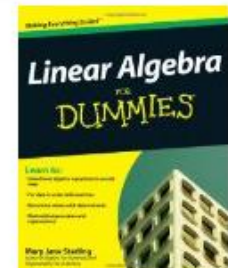
Linear Algebra: A Modern Introduction
› David Poole
Hardcover
★★★★☆ (41)
~~\$316.95~~ **\$289.88**



Linear Algebra
› G. E. Shilov
Paperback
★★★★☆ (34)
~~\$48.95~~ **\$12.65**



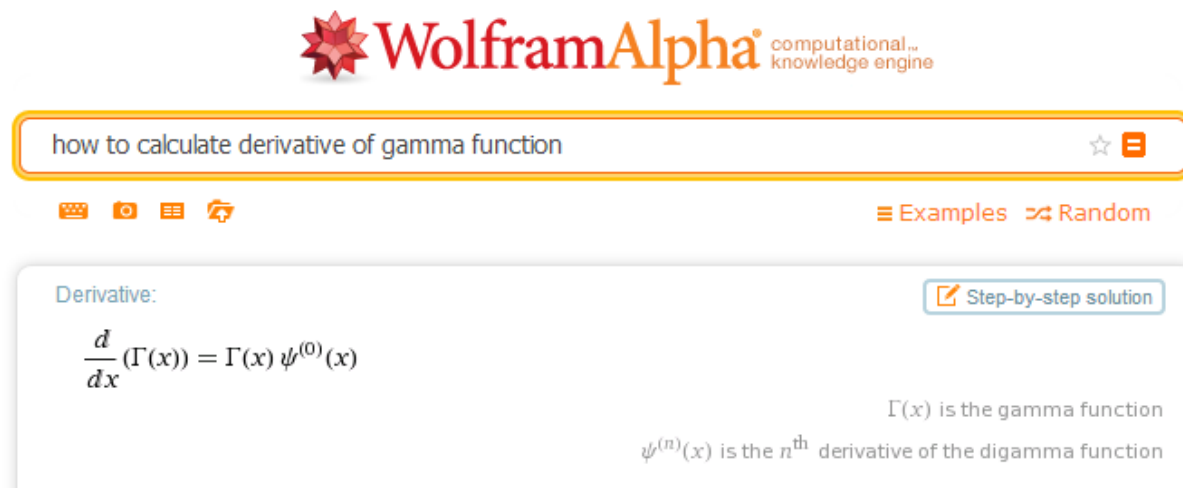
Introduction to Linear Algebra...
› Gilbert Strang
Hardcover
★★★★☆ (57)
~~\$87.50~~ **\$83.13**



Linear Algebra For Dummies
› Mary Jane Sterling
Paperback
★★★★☆ (29)
~~\$49.99~~ **\$16.23**

IR is not just about web search

- Web search is just one important area of information retrieval, but not all
- Information retrieval also includes
 - Question answering



The screenshot shows the WolframAlpha interface. At the top is the WolframAlpha logo with the tagline "computational... knowledge engine". Below the logo is a search bar containing the text "how to calculate derivative of gamma function". To the right of the search bar are icons for a star and a menu. Below the search bar are icons for keyboard, camera, list, and share. To the right of these icons are links for "Examples" and "Random". Below the search bar is a box containing the result. On the left of this box is the text "Derivative:". To the right is a button labeled "Step-by-step solution". The main result is the equation $\frac{d}{dx}(\Gamma(x)) = \Gamma(x) \psi^{(0)}(x)$. Below this equation, on the right, is the text " $\Gamma(x)$ is the gamma function". Below that, on the left, is the text " $\psi^{(n)}(x)$ is the n^{th} derivative of the digamma function".

WolframAlpha computational... knowledge engine

how to calculate derivative of gamma function

Examples Random

Derivative:

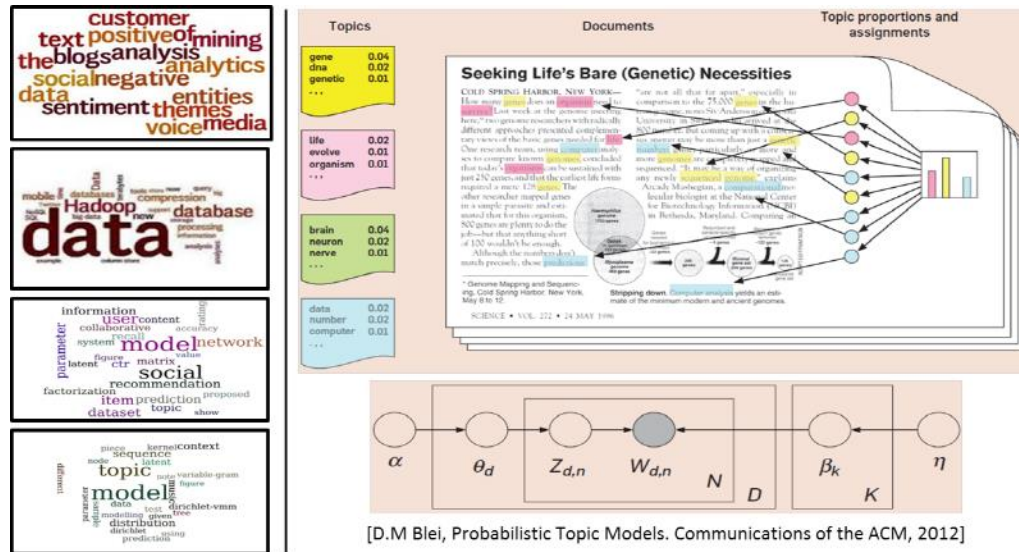
$\frac{d}{dx}(\Gamma(x)) = \Gamma(x) \psi^{(0)}(x)$

$\Gamma(x)$ is the gamma function

$\psi^{(n)}(x)$ is the n^{th} derivative of the digamma function

IR is not just about web search

- Web search is just one important area of information retrieval, but not all
- Information retrieval also includes
 - Text mining



[D.M Blei, Probabilistic Topic Models. Communications of the ACM, 2012]

IR is not just about web search

- Web search is just one important area of information retrieval, but not all
- Information retrieval also includes
 - Online advertising

The image is a screenshot of a Google search results page for the query "health care". The search bar at the top shows the query and a microphone icon. Below the search bar, there are tabs for "Web", "News", "Images", "Maps", "Books", and "More", along with a "Search tools" link. The search results indicate "About 782,000,000 results (0.45 seconds)".

The results are divided into organic search results and paid advertisements. The organic results include:

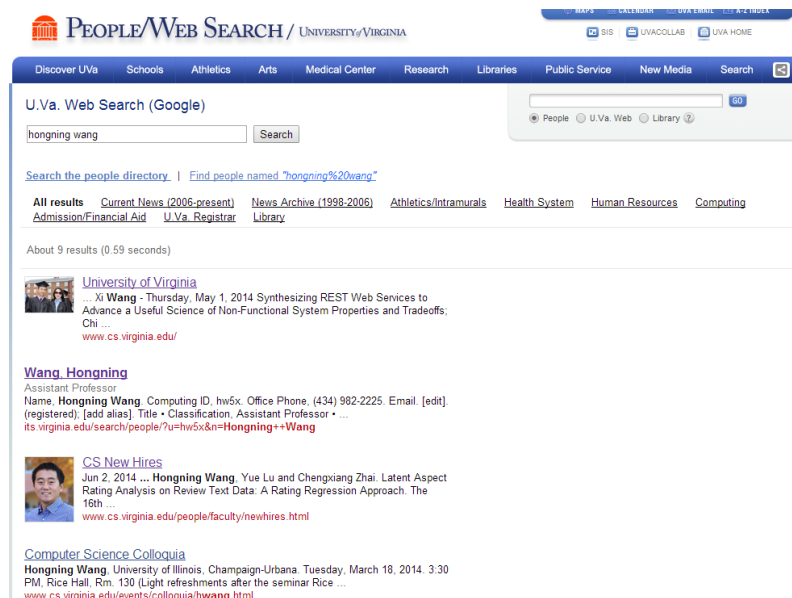
- HealthCare.gov: Health Insurance Marketplace, Affordable ...**
https://www.healthcare.gov/ HealthCare.gov
Learn how the health care law affects you at HealthCare.gov. The official site of the Health Insurance Marketplace. See your health insurance choices.
- Individuals & Families**
Individuals & Families. Still need health coverage? You can get ...
- Log In**
Marketplace Account Registration - Log In. New to HealthCare.gov ...
- More results from healthcare.gov »**
- Health Insurance Marketplace**
01. Apr. How to use your new Marketplace coverage ... Loss of ...
- Special Enrollment**
Medicaid - Qualifying Life Event - Complex Case - Exemptions

The paid advertisements are highlighted with red dashed boxes and include:

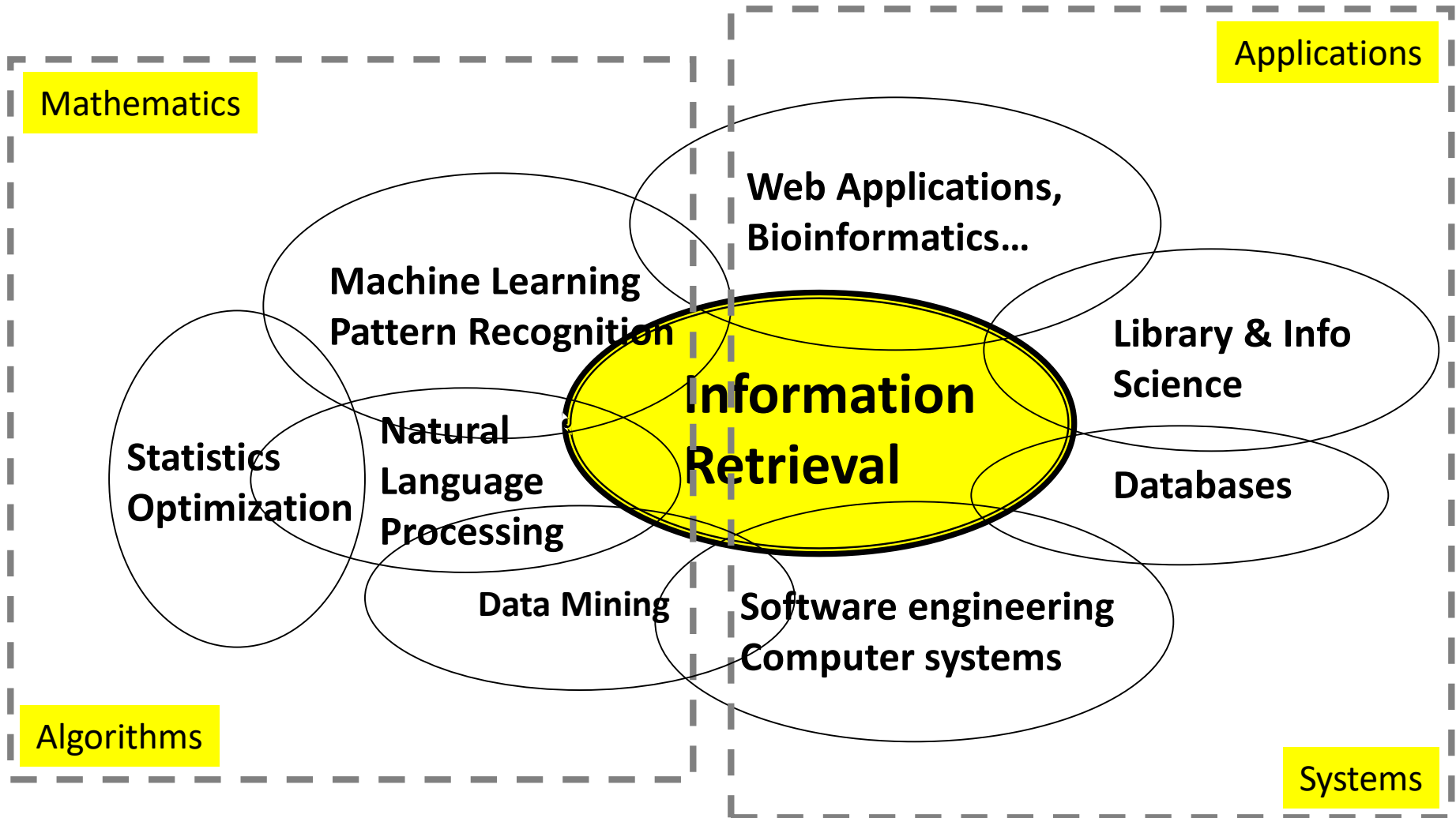
- Need Health Insurance? - MolinaHealthcare.com**
Ad www.molinahealthcare.com/ (877) 751-0665
Do You Qualify for Healthcare Reform Coverage? Find Out Now.
- Cheap Health Insurance - Only Takes A Few Minutes**
Ad www.healthquotejunction.com/ (888) 699-8397
Rates Starting Around \$100/mo.
- Low Cost Health Insurance - IndividualHealthQuotes.com**
Ad www.individualhealthquotes.com/ (866) 406-0696
Blue Cross, Aetna, CIGNA & More! (Illinois Residents Only)
- \$19 Health Insurance**
adffordable-health-insurance-plans.org/
New 2014 Discounts. Save 55% - 75%.
Compare Affordable Plans Online!
- Christie Clinic**
www.christieclinic.com/
Quality healthcare for east central Illinois
- Obama Health Care**
www.obama-care.org/
See If You Are Eligible & Apply For The Obama Care Health Plan.
- Low Cost Health Insurance**
www.directhealthinsurance.com/
Get Insured (Limited Time Only).
Compare Plans / Avoid Fees / Enroll
- Affordable Health Care**
www.hstinsurancequotes.com/affordable
Affordable Health Insurance Plans.
2014 Pricing Options - Free Quotes!

IR is not just about web search

- Web search is just one important area of information retrieval, but not all
- Information retrieval also includes
 - Enterprise search: web search + desktop search



Related Areas



IR v.s. DBs

- Information Retrieval:
 - Unstructured data
 - Semantics of object are subjective
 - Simple key word queries
 - Relevance-drive retrieval
 - Effectiveness is primary issue, though efficiency is also important
- Database Systems:
 - Structured data
 - Semantics of each object are well defined
 - Structured query languages (e.g., SQL)
 - Exact retrieval
 - Emphasis on efficiency

IR and DBs are getting closer

- IR => DBs

- Approximate search is available in DBs
- Eg. in MySQL

```
mysql> SELECT * FROM articles  
-> WHERE MATCH (title,body)  
AGAINST ('database');
```

- DBs => IR

- Use information extraction to convert unstructured data to structured data
- Semi-structured representation: XML data; queries with structured information

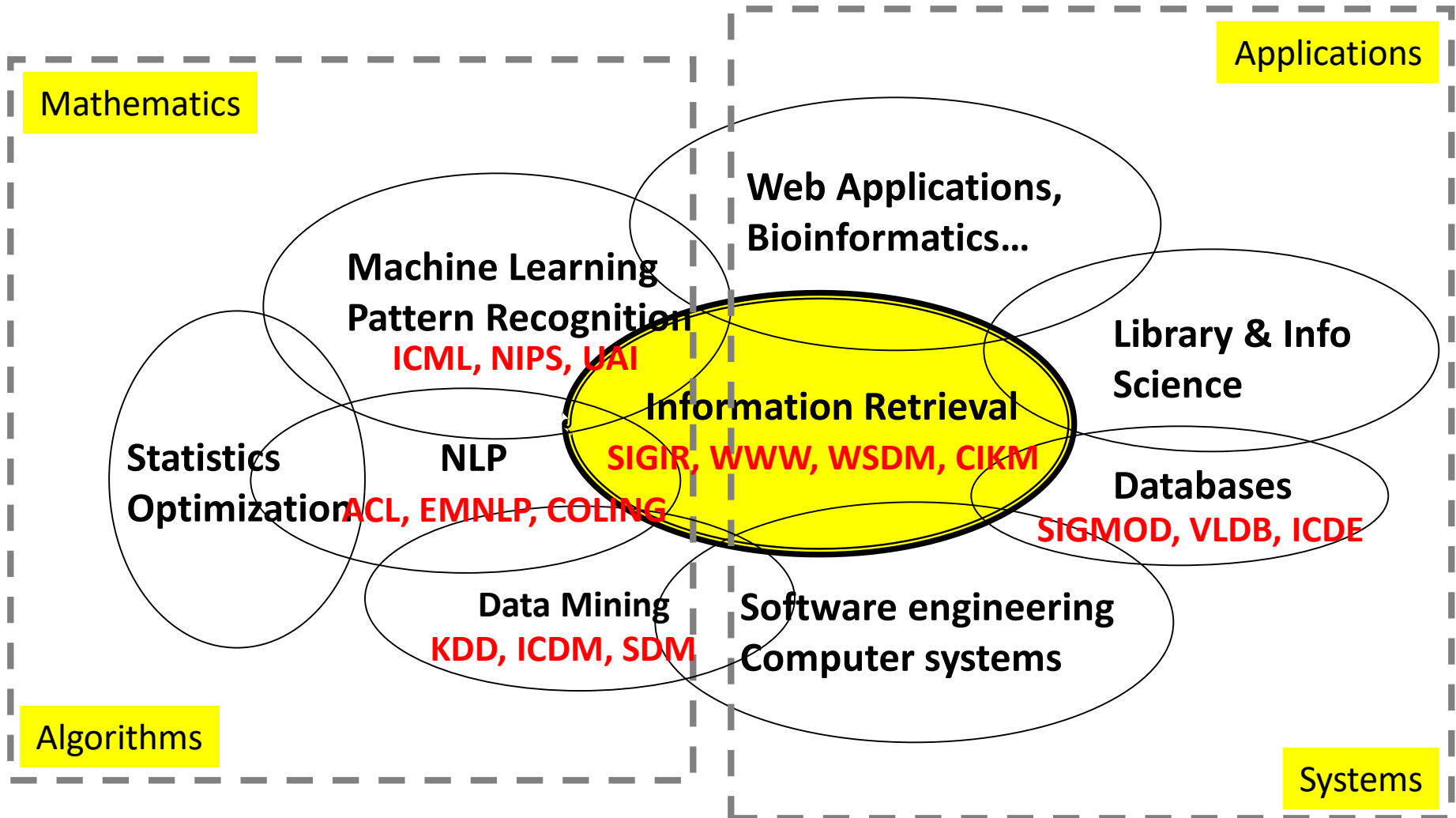
IR v.s. NLP

- Information retrieval
 - Computational approaches
 - Statistical (shallow) understanding of language
 - Handle large scale problems
- Natural language processing
 - Cognitive, symbolic and computational approaches
 - Semantic (deep) understanding of language
 - (often times) small scale problems

IR and NLP are getting closer

- IR => NLP
 - Larger data collections
 - Scalable/robust NLP techniques, e.g., translation models
- NLP => IR
 - Deep analysis of text documents and queries
 - Information extraction for structured IR tasks

What to read?



- Find more on course website for resource

Notes

- IR originates from library science for handling unstructured data
- IR has many important application areas, e.g., web search, recommendation, and question answering
- IR is a highly interdisciplinary area with DBs, NLP, ML, HCI